Homework 3

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

In each of Problems 1 and 2, transform the given initial value problem into an equivalent problem with the initial point at the origin

1. Exercise §2.8 #1.

$$dy/dt = t^2 + y^2, \qquad y(1) = 2$$

$u = t - 1 \qquad v = y - 2$

$du = dt \qquad dv = dy$

$\frac{dy}{dt} = \frac{dv}{du}$

$$\frac{dv}{du} = (u+1)^2 + (v+2)^2$$

$v(0) = 0$

2. Exercise §2.8 #2.

$$dy/dt = 1 - y^3, \qquad y(-1) = 3$$

$u = t + 1 \qquad v = y - 3$

$du = dt \qquad dv = dy$

$\frac{dy}{dt} = \frac{dv}{du}$

$$\frac{dv}{du} = 1 - (v+3)^3$$

$v(0) = 0$

In each of Problems 4 and 5, let $\phi_0(t) = 0$ and define $\{\phi_n(t)\}$ by the method of successive approximations
(a) Determine $\{\phi_n(t)\}$ for an arbitrary value of n.
(b) Plot $\{\phi_n(t)\}$ for $n = 1, ..., 4$. Observe whether the iterates appear to be converging.
(c) Express $\lim_{n\to\infty} \phi_n(t) = \phi(t)$ in terms of elementary functons; that is, solve the given intial value problem.
(d) Plot $|\phi(t) - \phi_n(t)|$ for $n = 1, ..., 4$. For each of $\phi_1(t), ..., \phi_4(t)$, estimate the interval in which it is a reasonably good approximation to the actual solution.

3. Exercise §2.8 #4.

$$y' = -y - 1, \qquad y(0) = 0$$

a) $y = -\int_0^t (y+1)\,ds = \varphi(t)$

b) $\varphi_1(t) = -\int_0^t (\varphi_0(t)+1)\,ds$

$= -\int_0^t ds = -t$

$\varphi_2(t) = -\int_0^t (\varphi_1(t)+1)\,ds = \frac{1}{2}t^2 - t$

$\varphi_3(t) = -\int_0^t (\varphi_2(t)+1)\,ds = -\frac{1}{6}t^3 + \frac{1}{2}t^2 - t$

$\varphi_4(t) = -\int_0^t (\varphi_3(t)+1)\,ds = \frac{1}{24}t^4 - \frac{1}{6}t^3 + \frac{1}{2}t^2 - t$

$\varphi_n(t) = -\int_0^t (\varphi_{n-1}(s)+1)\,ds$

$n : \{n \in \mathbb{Z}^+\}$

$\varphi_0(t) = 0$

proof of pattern see page 3.

actual graphical plots also see page 3

c) $\lim_{n\to\infty} \varphi_n(t) = \frac{(-t)^1}{1!} + \frac{(-t)^2}{2!} + \cdots + \frac{(-t)^n}{n!} = \sum_{n=1}^{\infty} \frac{(-t)^n}{n!} = e^{-t} - 1 = y(t) = \varphi(t)$

d) see page 3.

4. Exercise §2.8 #5.

$$y' = -y/2 + t, \qquad y(0) = 0$$

a) $y(t) = \int_0^t (-\frac{1}{2}y + s)\,ds = \varphi(t)$

b) $\varphi_1(t) = \int_0^t (-\frac{1}{2}\varphi_0(s) + s)\,ds = \frac{1}{2}t^2$

$\varphi_2(t) = \int_0^t (-\frac{1}{2}\varphi_1(s) + s)\,ds = -\frac{1}{12}t^3 + \frac{1}{2}t^2$

$\varphi_3(t) = \int_0^t (-\frac{1}{2}\varphi_2(s) + s)\,ds = \frac{1}{96}t^4 - \frac{1}{12}t^3 + \frac{1}{2}t^2$

$\varphi_4(t) = \int_0^t (-\frac{1}{2}\varphi_3(s) + s)\,ds = \frac{-1}{960}t^5 + \frac{1}{96}t^4 - \frac{1}{12}t^3 + \frac{1}{2}t^2$

$\varphi_n(t) = \sum_{k=1}^{n} \frac{(-t)^{k+1}}{2^{k-1}(k+1)!}$

$\varphi_n(t) = \int_0^t (-\frac{1}{2}\varphi_{n-1}(s) + s)\,ds$

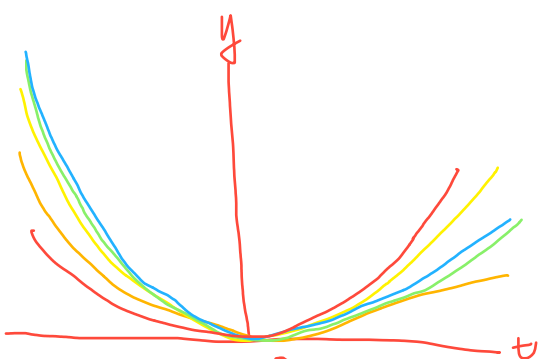
$\varphi_0(t) = 0$, $n : \{n \in \mathbb{Z}^+\}$

I.D.E $\varphi_{n+1}(t) = \int_0^t (-\frac{1}{2}\sum_{k=1}^{n} \frac{(-s)^{k+1}}{2^{k-1}(k+1)!} + s)\,ds$

$= -\frac{1}{2}\frac{-1}{k+2}\sum_{k=1}^{n} \frac{(-s)^{k+2}}{2^{k-1}(k+1)!} + \frac{1}{2}s^2$

$= \sum_{k=2}^{n+1} \frac{(-s)^{k+2}}{2^k(k+2)!} + \frac{1}{2}s^2$

$= \sum_{k=1}^{n+1} \frac{(-s)^{k+2}}{2^k(k+2)!}$ = expected $\varphi_{n+1}(t)$ ☒



$-\varphi(t)$
$-\varphi_4(t)$
$-\varphi_3(t)$
$-\varphi_2(t)$
$-\varphi_1(t)$

converging

drawn relative to each other with scaled horizontal axis

more see page 4

3. b) cont. proof by induction

assume: $\varphi_n(t) = \sum_{k=1}^{n} \frac{(-t)^k}{k!}$

then: $\varphi_{n+1}(t) = \sum_{k=1}^{n+1} \frac{(-t)^k}{k!}$

Q.E.D. $\varphi_{n+1}(t) = -\int_0^t (\varphi_n(s)+1)\,ds$

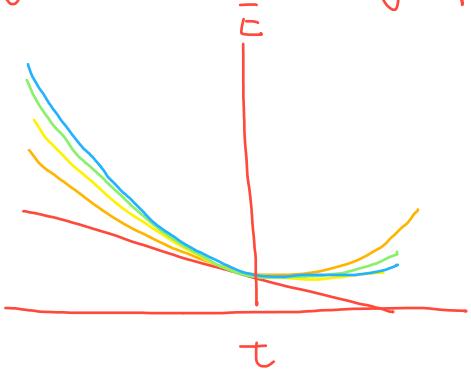
$= -\int_0^t \sum_{k=1}^{n} \frac{(-s)^k}{k!}\,ds - t$

$= \frac{1}{k+1} \sum_{k=1}^{n} \frac{(-t)^{k+1}}{k!} - t$

$= \sum_{k=2}^{n+1} \frac{(-t)^k}{k!} - t$

$= \sum_{k=1}^{n+1} \frac{(-t)^k}{k!} = \varphi_{n+1}(t)$ ▨

cont. graphical plots graphed by Desmos



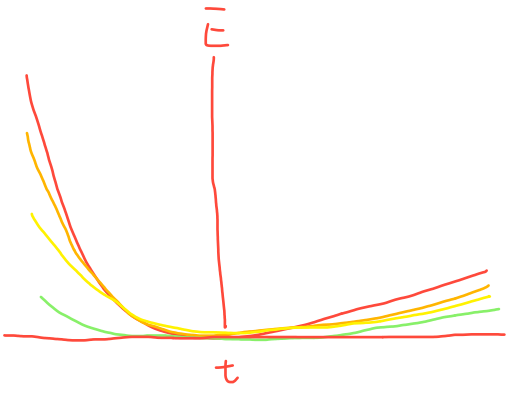
- : $\varphi(t)$
- : $\varphi_4(t)$
- : $\varphi_3(t)$
- : $\varphi_2(t)$
- : $\varphi_1(t)$

converging

d) $E_n(t) = \left| e^{-t} - 1 - \sum_{k=1}^{n} \frac{(-t)^k}{k!} \right|$

taking $E_n(t) < 0.1$ as a good aprox:

φ_1	$I: \{-0.42 < t < 0.48\}$
φ_2	$I: \{-0.79 < t < 0.91\}$
φ_3	$I: \{-1.17 < t < 1.32\}$
φ_4	$I: \{-1.55 < t < 1.73\}$



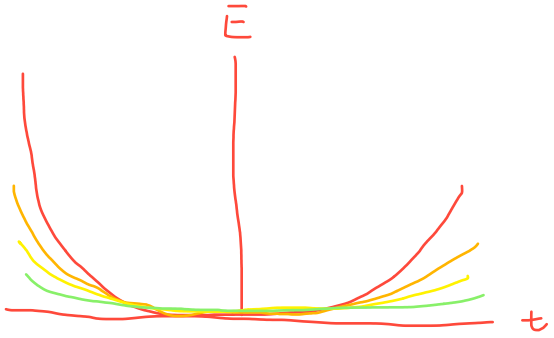
- : $|\varphi(t) - \varphi_4(t)|$
- : $|\varphi(t) - \varphi_3(t)|$
- : $|\varphi(t) - \varphi_2(t)|$
- : $|\varphi(t) - \varphi_1(t)|$

4.c) $\varphi_n(t) = \sum_{k=1}^{n} \frac{(-t)^{k+1}}{2^{k-1}(k+1)!} = \sum_{k=2}^{n} \frac{4(-t)^k}{2^k\, k!} = 4\sum_{k=2}^{n} \frac{(-\frac{1}{2}t)^k}{k!}$

$\lim_{n\to\infty} \varphi_n(t) = 4\sum_{k=2}^{\infty} \frac{(-\frac{1}{2}t)^k}{k!} = 4\left(e^{-\frac{1}{2}t} - 1 + \frac{1}{2}t\right) = \varphi(t) = y(t)$

d) $E_n(t) = \left| 4\left(e^{-\frac{1}{2}t} - 1 + \frac{1}{2}t\right) - 4\sum_{k=2}^{n} \frac{(-\frac{1}{2}t)^k}{k!} \right|$

taking $E_n(t) < 0.1$ as a good aprox:



- : $|\varphi(t) - \varphi_4(t)|$
- : $|\varphi(t) - \varphi_3(t)|$
- : $|\varphi(t) - \varphi_2(t)|$
- : $|\varphi(t) - \varphi_1(t)|$

φ_1 $I : \{-1.02 < t < 1.11\}$

φ_2 $I : \{-1.68 < t < 1.84\}$

φ_3 $I : \{-2.39 < t < 2.59\}$

φ_4 $I : \{-3.11 < t < 3.36\}$